

USER MANUAL

Introduction

This user manual explains the installation procedure and execution steps for the proposed project titled **"Automated Brain Tumor Detection from MRI Images Using Transfer Learning with ResNet50."** The project is implemented using Python, TensorFlow/Keras, Google Colab, and deep learning techniques for automated MRI brain tumor classification.

System Requirements

Hardware Requirements

- Intel Core i5 Processor or higher
 - Minimum 8 GB RAM
 - Stable Internet Connection
 - GPU recommended for faster training
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Software Requirements

- Google Colab / Jupyter Notebook
 - Python 3.9 or above
 - Google Drive Account
 - Required Python Libraries:
 - TensorFlow
 - Keras
 - NumPy
 - Pandas
 - Matplotlib
 - OpenCV
 - Scikit-learn
 - Seaborn
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Installation Procedure

Step 1: Open Google Colab

Open Google Colab using the following link:

Step 2: Upload Project Files

Upload the following files to Google Drive:

- MRI Image Dataset ZIP File
 - CSV Dataset File
 - Python Notebook / Colab File
 - Trained Model File
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Step 3: Mount Google Drive

Run the following command to connect Google Drive:

```
``python id="0i4mk9" from google.colab import drive drive.mount('/content/drive')
```

```
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## Step 4: Extract Dataset  
  
Extract the MRI image dataset ZIP file:  
  
``python id="1mqo8p"  
import zipfile  
  
zip_path = "/content/drive/MyDrive/Images.zip"  
extract_path = "/content/images"  
  
with zipfile.ZipFile(zip_path, 'r') as zip_ref:  
    zip_ref.extractall(extract_path)  
  
print("Unzipped successfully")
```

Step 5: Install Required Libraries

Run the following command:

```
``python id="3r0xpu" pip install tensorflow keras numpy pandas matplotlib opencv-python scikit-learn  
seaborn
```

```

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# Running the Project

## Step 1: Import Required Libraries

Execute all import statements including TensorFlow, NumPy, Pandas, OpenCV, and
Keras libraries.

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## Step 2: Load Dataset

Load the CSV dataset and MRI images from the extracted folder.

The dataset contains:

- MRI Brain Images
- Tumor and Non-Tumor Classes
- Texture and Statistical Features

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## Step 3: Image Preprocessing

The framework automatically performs:

- Image resizing (224 × 224)
- Image normalization
- Data augmentation
- Feature preparation

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## Step 4: Apply K-Means Clustering

K-Means clustering is applied for feature grouping and similarity analysis
before classification.

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## Step 5: Train ResNet50 Model

The transfer learning model is initialized using pretrained ResNet50 weights.

Training configuration:

```

- Batch Size : 16
- Epochs : 10
- Optimizer : Adam
- Learning Rate : 0.0001

Step 6: Model Evaluation

The framework evaluates the model using:

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix

The proposed framework achieved approximately **90.84%** classification accuracy.

Tumor Prediction Procedure

Step 1: Enter MRI Image Number

Provide the image number during execution.

Example:

```
```python id="9lpt5z"  
Enter image number (e.g., 1, 25, 300): 10
```

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## Step 2: Prediction Output

The framework displays:

- Tumor Detected / Non-Tumor
  - Prediction Confidence Score
  - MRI Image Visualization
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# Grad-CAM Visualization

The proposed system also generates Grad-CAM heatmap visualization for highlighting important tumor regions in MRI images.

The visualization helps improve interpretability of the deep learning prediction results.

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## Output Generated

The proposed framework generates:

- Tumor prediction result
  - Accuracy graph
  - Loss graph
  - Confusion matrix
  - Classification report
  - Grad-CAM visualization
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## Precautions

- Ensure correct dataset path before execution.
  - Install all required libraries properly.
  - Use GPU runtime in Google Colab for faster execution.
  - Maintain correct image naming format inside dataset folders.
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## Conclusion

The proposed framework provides an efficient and automated solution for brain tumor detection using MRI images. The integration of transfer learning with ResNet50 improves classification accuracy and supports reliable medical image analysis.